Mid-term PROJECT

I. Projects

- ***1. Linear Regression: MovieLens 20M Dataset***
  - URL: https://grouplens.org/datasets/movielens/20m/
- ***2. Classification: Instacart Market Basket Analysis***
  - URL: https://www.kaggle.com/datasets/psparks/instacart-market-basket-analysis
- ***3. Clustering: HR Analytics Case Study***
  - URL: https://www.kaggle.com/datasets/vjchoudhary7/hr-analytics-case-study
- ***4. Classification: Employee Churn Prediction (Relational)***
  - URL: https://www.kaggle.com/datasets/vjchoudhary7/hr-analytics-case-study
- ***5. Linear Regression: Brazilian E-Commerce Public Dataset by Olist***
  - URL: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
- ***6. Classification: Book Recommendation Dataset***
  - URL: https://www.kaggle.com/datasets/arashnic/book-recommendation-dataset
- ***7. Clustering: Google Play Store Apps***
  - URL: https://www.kaggle.com/datasets/lava18/google-play-store-apps
- ***8. Classification: IEEE-CIS Fraud Detection***
  - URL: https://www.kaggle.com/c/ieee-fraud-detection/data
- ***9. Clustering: Retailrocket recommender system dataset***
  - URL: https://www.kaggle.com/datasets/retailrocket/ecommerce-dataset
- ***10. Clustering: Grocery Store Dataset (Transaction & Product)***
  - URL: https://www.kaggle.com/datasets/frtgnn/dunnhumby-the-complete-journey

II. Report deliverables ans requirements

To complete the project, each group must submit the following items:

1. Comprehensive Analysis Report (PDF File)

- **Format**: A single PDF report documenting the entire analysis process.
- **Technical Recommendation**: The use of LaTeX is highly encouraged for professional formatting, clear structure and precise rendering of mathematical formulas and tables.
- **Core Content**:

- A full explanation of every major step, from problem definition to final results.
- Visualizations and Outputs: Inclusion of illustrative output images, analysis charts, and figures with detailed interpretations.
- Evaluation: A deep dive into model performance and comparison based on specific criteria (e.g., Accuracy, Baseline comparison, Confusion Matrix, etc.).

2. Source Code and Version Control (GitHub)

- **Requirements**: One .ipynb notebook that runs from beginning to end on Colab or Jupyter.
- **Documentation**: The code must contain clear comments and brief explanations.
- **Repository**: A public GitHub repository link must be provided.

3. Presentation Slides

- **Length**: Between 10 to 15 slides.
- **Content**: A concise summary of the problem, dataset, methodology, results, and conclusions.

III. Suggestions on how to do the exercise

1. Data Inspection

- **Check for consistency**: Do the files have the same number of columns and the same column names? (Example: monthly sales data).
- **Identify relationships**: If the files contain different information (e.g., customers.csv and orders.csv), find the common columns (Primary Key/Foreign Key) to connect them later.
- **Check encoding**: Determine if the files are UTF-8 or another format to avoid font errors during loading.

2. Data Loading and Integration

Instead of reading each file manually, use code to automate the process:

- **Concatenation**: Used when files have identical structures.
- **Merging/Joining**: Used when you need to supplement information from different tables into a main table.

3. Data Cleaning

- **Handle missing values**: Decide whether to remove or fill them (imputation) using mean/median values.
- **Remove duplicates**: This is especially likely to happen when merging multiple files.
- **Standardize data types**: Ensure date columns are datetime and revenue columns are float or int.

4. Exploratory Data Analysis (EDA)

In data analysis, EDA is not merely about glancing at numbers but is a process of "interrogating" the data to uncover hidden patterns and intrinsic structures. The core objective of EDA is to test hypotheses, detect outliers, and assess the relationships between data fields through statistical metrics or a correlation matrix. Understanding these relationships helps you perform the most critical step: Feature Selection.

To serve specific problems, you need to define the role of each data field:

- **Regression and Classification**: You need to identify the target variable (Output/Target) – usually the result you want to find (such as house prices or a "yes/no" label) – and the independent variables (Input/Features), which are factors that have a direct impact or the strongest correlation with that target.
- **Clustering**: Unlike the two types above, EDA here helps you understand the data distribution to select input fields with the best group separation potential (such as shopping behavior or age) without needing to pre-define an output variable, as the goal is to find naturally existing group structures within the dataset.

Choosing the right input variables not only helps the model run faster but also avoids "noise", ensuring higher reliability in your analytical results.

5. Implementing Machine Learning Algorithms

Depending on the problem objective, choose the appropriate group of algorithms:

- **Regression**: Linear Regression, Random Forest Regressor, XGBoost.
- **Classification**: Logistic Regression, Decision Tree, Support Vector Machine (SVM), Neural Networks.
- **Clustering**: K-Means, DBSCAN, Hierarchical Clustering.

6. Model Evaluation

This is the step to determine if the model is truly effective:

- **Establish a Baseline**: Always start with the simplest model (e.g., using the mean value or a simple Linear model) as a benchmark for comparison. If a complex model cannot beat the Baseline, you need to re-examine the data.
- **Use Metrics**:

- ***Regression***: MAE (Mean Absolute Error), RMSE (Root Mean Square Error), R^2.
- ***Classification***: Accuracy, Precision, Recall, F1-Score and the AUC-ROC curve.
- ***Clustering***: Silhouette Score (measure of cluster separation), Elbow Method.

7. Conclusion

In this part, the group must provide a comprehensive summary that synthesizes the entire analytical journey. This includes highlighting the main insights extracted from the data and identifying the best-performing model based on the evaluation metrics. Beyond the technical results, the group should discuss the practical implications of these findings, how they can be applied to solve real-world problems or inform business decisions. To maintain transparency and academic rigor, it is essential to address the limitations of the project, such as data constraints or algorithmic boundaries. Finally, the conclusion should outline possible future improvements, suggesting how the model or analysis could be refined or expanded in subsequent iterations.